

Mathematics for Machine Learning

Vazgen Mikayelyan



Sequence of Functions

Sequence of Functions

Let $f_n : X \rightarrow \mathbb{R}, n \in \mathbb{N}$ be a sequence of functions.

Sequence of Functions

Let $f_n : X \rightarrow \mathbb{R}, n \in \mathbb{N}$ be a sequence of functions.

Definition

We will say that f_n converges to f everywhere on X , if
 $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ *for all $x \in X$.*

Sequence of Functions

Let $f_n : X \rightarrow \mathbb{R}, n \in \mathbb{N}$ be a sequence of functions.

Definition

We will say that f_n converges to f everywhere on X , if
$$\lim_{n \rightarrow \infty} f_n(x) = f(x) \text{ for all } x \in X.$$

Definition

We will say that f_n converges to f uniformly on X , if for all $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ holds $|f_n(x) - f(x)| < \varepsilon$ for all $x \in X$. We will write $f_n(x) \rightrightarrows f(x), x \in X$.

Sequence of Functions

Let $f_n : X \rightarrow \mathbb{R}, n \in \mathbb{N}$ be a sequence of functions.

Definition

We will say that f_n converges to f everywhere on X , if
$$\lim_{n \rightarrow \infty} f_n(x) = f(x) \text{ for all } x \in X.$$

Definition

We will say that f_n converges to f uniformly on X , if for all $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ holds $|f_n(x) - f(x)| < \varepsilon$ for all $x \in X$. We will write $f_n(x) \rightrightarrows f(x), x \in X$.

Theorem

f_n converges uniformly on X if and only if for all $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that for all $n \geq n_0, m \in \mathbb{N}$ holds $|f_{n+m}(x) - f_n(x)| < \varepsilon$ for all $x \in X$.

Series of Functions

Series of Functions

Let $u_n : X \rightarrow \mathbb{R}, n \in \mathbb{N}$ be a sequence of functions.

Series of Functions

Let $u_n : X \rightarrow \mathbb{R}, n \in \mathbb{N}$ be a sequence of functions.

Definition

We will say that the series $\sum_{n=1}^{\infty} u_n(x)$ converges to f everywhere on X , if the sequence $f_n(x) = \sum_{k=1}^n u_k(x)$ converges to f everywhere on X .

Series of Functions

Let $u_n : X \rightarrow \mathbb{R}, n \in \mathbb{N}$ be a sequence of functions.

Definition

We will say that the series $\sum_{n=1}^{\infty} u_n(x)$ converges to f everywhere on X , if the sequence $f_n(x) = \sum_{k=1}^n u_k(x)$ converges to f everywhere on X .

Definition

We will say that the series $\sum_{n=1}^{\infty} u_n(x)$ converges to f uniformly on X , if the sequence $f_n(x) = \sum_{k=1}^n u_k(x)$ converges to f uniformly on X .

Theorem

If there exists a sequence of real numbers λ_n such that $|u_n(x)| \leq \lambda_n$ for all $x \in X$ and $n \in \mathbb{N}$ and $\sum_{n=1}^{\infty} \lambda_n < +\infty$, then $\sum_{n=1}^{\infty} u_n(x)$ is uniformly convergent.

Theorem

If there exists a sequence of real numbers λ_n such that $|u_n(x)| \leq \lambda_n$ for all $x \in X$ and $n \in \mathbb{N}$ and $\sum_{n=1}^{\infty} \lambda_n < +\infty$, then $\sum_{n=1}^{\infty} u_n(x)$ is uniformly convergent.

Example

Prove that the series $\sum_{n=1}^{\infty} \frac{\sin nx}{1+n^2}$ is uniformly convergent on $\mathbb{R}$.

Theorem

If $f_n \in C(X)$, $n \in \mathbb{N}$ and $f_n(x) \Rightarrow f(x)$, $x \in X$, then $f \in C(X)$.

Sequence of Functions

Theorem

If $f_n \in C(X)$, $n \in \mathbb{N}$ and $f_n(x) \Rightarrow f(x)$, $x \in X$, then $f \in C(X)$.

Theorem

If $f_n(x) \Rightarrow f(x)$, $x \in X$ and $\lim_{x \rightarrow a} f_n(x) = c_n$ for all $n \in \mathbb{N}$, then there exists finite limits $\lim_{n \rightarrow \infty} c_n = c$ and $\lim_{x \rightarrow a} f(x) = c$. In other words

$$\lim_{x \rightarrow a} \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \lim_{x \rightarrow a} f_n(x).$$

Sequence of Functions

Theorem

If $f_n \in C[a, b]$, $n \in \mathbb{N}$ and $f_n(x) \Rightarrow f(x)$, $x \in [a, b]$, then

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx$$

Sequence of Functions

Theorem

If $f_n \in C[a, b]$, $n \in \mathbb{N}$ and $f_n(x) \Rightarrow f(x)$, $x \in [a, b]$, then

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx$$

Theorem

If $f'_n \in C[a, b]$, $n \in \mathbb{N}$, $f_n(x) \rightarrow f(x)$, $x \in [a, b]$ and $f'_n(x) \Rightarrow g(x)$, $x \in [a, b]$, then $f'(x) = g(x)$.

Sequence of Functions

Theorem

If $f_n \in C[a, b]$, $n \in \mathbb{N}$ and $f_n(x) \Rightarrow f(x)$, $x \in [a, b]$, then

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx$$

Theorem

If $f'_n \in C[a, b]$, $n \in \mathbb{N}$, $f_n(x) \rightarrow f(x)$, $x \in [a, b]$ and $f'_n(x) \Rightarrow g(x)$, $x \in [a, b]$, then $f'(x) = g(x)$.

Theorem

If $f \in C[a, b]$, then there exists a sequence of polynomials P_n , such that

$$P_n(x) \Rightarrow f(x), x \in [a, b].$$

Definition

Power series are the series of the form $\sum_{n=0}^{\infty} a_n (x - x_0)^n$.

Power Series

Definition

Power series are the series of the form $\sum_{n=0}^{\infty} a_n (x - x_0)^n$.

Theorem

If $R = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}}$, then the series $\sum_{n=0}^{\infty} a_n x^n$ is convergent if $|x| < R$ and is divergent for $|x| > R$.

Power Series

Definition

Power series are the series of the form $\sum_{n=0}^{\infty} a_n (x - x_0)^n$.

Theorem

If $R = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}}$, then the series $\sum_{n=0}^{\infty} a_n x^n$ is convergent if $|x| < R$ and is divergent for $|x| > R$.

Theorem

If $R > 0$ then $\sum_{n=0}^{\infty} a_n x^n \in C(-R, R)$.

Theorem

If $R \in (0, \infty)$ and the series $\sum_{n=0}^{\infty} a_n R^n$ is convergent then the series

$\sum_{n=0}^{\infty} a_n x^n$ is uniformly convergent on $[0, R]$.

Theorem

If $f(x) = \sum_{n=0}^{\infty} a_n x^n$ and $R > 0$, then $f \in \mathcal{R}[0, a]$ for all $a \in (0, R)$ and

$$\int_0^a f(x) dx = \sum_{n=0}^{\infty} \frac{a_n}{n+1} a^{n+1}$$

Theorem

If $f(x) = \sum_{n=0}^{\infty} a_n x^n$ and $R > 0$, then $f \in \mathcal{R}[0, a]$ for all $a \in (0, R)$ and

$$\int_0^a f(x) dx = \sum_{n=0}^{\infty} \frac{a_n}{n+1} a^{n+1}$$

Theorem

If $f(x) = \sum_{n=0}^{\infty} a_n x^n$ and $R > 0$, then f is differentiable in $(-R, R)$ and

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

Convergence of Random Variables

Convergence of Random Variables

Definition

A sequence of random variables X_n converges in distribution to X , if their CDFs F_n converge to the CDF of X (F) at any point of continuity of F .

Convergence of Random Variables

Definition

A sequence of random variables X_n converges in distribution to X , if their CDFs F_n converge to the CDF of X (F) at any point of continuity of F .

Definition

A sequence of random variables X_n converges to X almost surely, if

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} X_n = X \right) = 1.$$

Convergence of Random Variables

Definition

A sequence of random variables X_n converges in probability to X , if for any $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| \geq \varepsilon) = 0.$$

Convergence of Random Variables

Definition

A sequence of random variables X_n converges in probability to X , if for any $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| \geq \varepsilon) = 0.$$

Definition

A sequence of random variables X_n converges to X in the r -th mean ($r > 0$), if $\mathbb{E}[|X_n|^r] < +\infty$ for all $n \in \mathbb{N}$, $\mathbb{E}[|X|^r] < +\infty$ and

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n - X|^r] = 0.$$

Examples

Let X_n be an exponential random variable with parameter n , i.e. CDF of X_n is

$$F_n(x) = \begin{cases} 1 - e^{-nx}, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0. \end{cases}$$

Examples

Let X_n be an exponential random variable with parameter n , i.e. CDF of X_n is

$$F_n(x) = \begin{cases} 1 - e^{-nx}, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0. \end{cases}$$

Then for all $\varepsilon > 0$ we obtain

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n| \geq \varepsilon) =$$

Examples

Let X_n be an exponential random variable with parameter n , i.e. CDF of X_n is

$$F_n(x) = \begin{cases} 1 - e^{-nx}, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0. \end{cases}$$

Then for all $\varepsilon > 0$ we obtain

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n| \geq \varepsilon) = \lim_{n \rightarrow \infty} (1 - F_n(\varepsilon)) =$$

Examples

Let X_n be an exponential random variable with parameter n , i.e. CDF of X_n is

$$F_n(x) = \begin{cases} 1 - e^{-nx}, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0. \end{cases}$$

Then for all $\varepsilon > 0$ we obtain

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n| \geq \varepsilon) = \lim_{n \rightarrow \infty} (1 - F_n(\varepsilon)) = \lim_{n \rightarrow \infty} e^{-n\varepsilon} =$$

Examples

Let X_n be an exponential random variable with parameter n , i.e. CDF of X_n is

$$F_n(x) = \begin{cases} 1 - e^{-nx}, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0. \end{cases}$$

Then for all $\varepsilon > 0$ we obtain

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n| \geq \varepsilon) = \lim_{n \rightarrow \infty} (1 - F_n(\varepsilon)) = \lim_{n \rightarrow \infty} e^{-n\varepsilon} = 0.$$

Examples

Let X_n be an exponential random variable with parameter n , i.e. CDF of X_n is

$$F_n(x) = \begin{cases} 1 - e^{-nx}, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0. \end{cases}$$

Then for all $\varepsilon > 0$ we obtain

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n| \geq \varepsilon) = \lim_{n \rightarrow \infty} (1 - F_n(\varepsilon)) = \lim_{n \rightarrow \infty} e^{-n\varepsilon} = 0.$$

So $X_n \xrightarrow{\mathbb{P}} 0$.

Convergence of Random Variables

Theorem

If $X_n \xrightarrow{(r)} X$, then $X_n \xrightarrow{\mathbb{P}} X$.

Convergence of Random Variables

Theorem

If $X_n \xrightarrow{(r)} X$, then $X_n \xrightarrow{\mathbb{P}} X$.

Theorem

If $X_n \xrightarrow{a.s.} X$, then $X_n \xrightarrow{\mathbb{P}} X$.

Convergence of Random Variables

Theorem

If $X_n \xrightarrow{(r)} X$, then $X_n \xrightarrow{\mathbb{P}} X$.

Theorem

If $X_n \xrightarrow{a.s.} X$, then $X_n \xrightarrow{\mathbb{P}} X$.

Theorem

If $X_n \xrightarrow{\mathbb{P}} X$ then there exists $k_n \in \mathbb{N}$ such that $X_{k_n} \xrightarrow{a.s.} X$.

Convergence of Random Variables

Theorem

If $X_n \xrightarrow{(r)} X$, then $X_n \xrightarrow{\mathbb{P}} X$.

Theorem

If $X_n \xrightarrow{a.s.} X$, then $X_n \xrightarrow{\mathbb{P}} X$.

Theorem

If $X_n \xrightarrow{\mathbb{P}} X$ then there exists $k_n \in \mathbb{N}$ such that $X_{k_n} \xrightarrow{a.s.} X$.

Theorem

If $X_n \xrightarrow{\mathbb{P}} X$, then $X_n \xrightarrow{D} X$.

Convergence of Random Variables

Theorem

If $X_n \xrightarrow{(r)} X$, then $X_n \xrightarrow{\mathbb{P}} X$.

Theorem

If $X_n \xrightarrow{a.s.} X$, then $X_n \xrightarrow{\mathbb{P}} X$.

Theorem

If $X_n \xrightarrow{\mathbb{P}} X$ then there exists $k_n \in \mathbb{N}$ such that $X_{k_n} \xrightarrow{a.s.} X$.

Theorem

If $X_n \xrightarrow{\mathbb{P}} X$, then $X_n \xrightarrow{D} X$.

Theorem

$X_n \xrightarrow{\mathbb{P}} c$ iff $X_n \xrightarrow{D} c$.

Examples

Let $\Omega = [0, 1]$, P is the Lebesgue measure and $\mathcal{F}$ is the σ -algebra of all measurable sets.

Examples

Let $\Omega = [0, 1]$, P is the Lebesgue measure and $\mathcal{F}$ is the σ -algebra of all measurable sets. Denote

$$X_n(\omega) = \begin{cases} e^n, & \text{if } 0 \leq \omega \leq \frac{1}{n}, \\ 0, & \text{if } \frac{1}{n} < \omega \leq 1. \end{cases}$$

Examples

Let $\Omega = [0, 1]$, P is the Lebesgue measure and $\mathcal{F}$ is the σ -algebra of all measurable sets. Denote

$$X_n(\omega) = \begin{cases} e^n, & \text{if } 0 \leq \omega \leq \frac{1}{n}, \\ 0, & \text{if } \frac{1}{n} < \omega \leq 1. \end{cases}$$

For any $\omega \in (0, 1]$ there exists $n_0 \in \mathbb{N}$ such that $X_n(\omega) = 0$ for $n \geq n_0$.

Examples

Let $\Omega = [0, 1]$, P is the Lebesgue measure and $\mathcal{F}$ is the σ -algebra of all measurable sets. Denote

$$X_n(\omega) = \begin{cases} e^n, & \text{if } 0 \leq \omega \leq \frac{1}{n}, \\ 0, & \text{if } \frac{1}{n} < \omega \leq 1. \end{cases}$$

For any $\omega \in (0, 1]$ there exists $n_0 \in \mathbb{N}$ such that $X_n(\omega) = 0$ for $n \geq n_0$. Therefore $X_n(\omega) \rightarrow 0$, for all $\omega \in (0, 1]$, i.e. $X_n \xrightarrow{a.s.} 0$.

Examples

Let $\Omega = [0, 1]$, P is the Lebesgue measure and $\mathcal{F}$ is the σ -algebra of all measurable sets. Denote

$$X_n(\omega) = \begin{cases} e^n, & \text{if } 0 \leq \omega \leq \frac{1}{n}, \\ 0, & \text{if } \frac{1}{n} < \omega \leq 1. \end{cases}$$

For any $\omega \in (0, 1]$ there exists $n_0 \in \mathbb{N}$ such that $X_n(\omega) = 0$ for $n \geq n_0$. Therefore $X_n(\omega) \rightarrow 0$, for all $\omega \in (0, 1]$, i.e. $X_n \xrightarrow{a.s.} 0$.

For any $\varepsilon > 0$ we have

$$\mathbb{P}(X_n \geq \varepsilon) \leq \frac{1}{n}, n \in \mathbb{N},$$

Examples

Let $\Omega = [0, 1]$, P is the Lebesgue measure and $\mathcal{F}$ is the σ -algebra of all measurable sets. Denote

$$X_n(\omega) = \begin{cases} e^n, & \text{if } 0 \leq \omega \leq \frac{1}{n}, \\ 0, & \text{if } \frac{1}{n} < \omega \leq 1. \end{cases}$$

For any $\omega \in (0, 1]$ there exists $n_0 \in \mathbb{N}$ such that $X_n(\omega) = 0$ for $n \geq n_0$. Therefore $X_n(\omega) \rightarrow 0$, for all $\omega \in (0, 1]$, i.e. $X_n \xrightarrow{a.s.} 0$.

For any $\varepsilon > 0$ we have

$$\mathbb{P}(X_n \geq \varepsilon) \leq \frac{1}{n}, n \in \mathbb{N},$$

therefore $X_n \xrightarrow{\mathbb{P}} 0$.

Examples

Let $\Omega = [0, 1]$, P is the Lebesgue measure and $\mathcal{F}$ is the σ -algebra of all measurable sets. Denote

$$X_n(\omega) = \begin{cases} e^n, & \text{if } 0 \leq \omega \leq \frac{1}{n}, \\ 0, & \text{if } \frac{1}{n} < \omega \leq 1. \end{cases}$$

For any $\omega \in (0, 1]$ there exists $n_0 \in \mathbb{N}$ such that $X_n(\omega) = 0$ for $n \geq n_0$. Therefore $X_n(\omega) \rightarrow 0$, for all $\omega \in (0, 1]$, i.e. $X_n \xrightarrow{a.s.} 0$.

For any $\varepsilon > 0$ we have

$$\mathbb{P}(X_n \geq \varepsilon) \leq \frac{1}{n}, n \in \mathbb{N},$$

therefore $X_n \xrightarrow{\mathbb{P}} 0$. Note that

$$\mathbb{E}[|X_n|^r] = \frac{e^{rn}}{n} \rightarrow +\infty, r > 0.$$

Examples

Let $\Omega = [0, 1]$, P is the Lebesgue measure and $\mathcal{F}$ is the σ -algebra of all measurable sets. Denote

$$X_n(\omega) = \begin{cases} e^n, & \text{if } 0 \leq \omega \leq \frac{1}{n}, \\ 0, & \text{if } \frac{1}{n} < \omega \leq 1. \end{cases}$$

For any $\omega \in (0, 1]$ there exists $n_0 \in \mathbb{N}$ such that $X_n(\omega) = 0$ for $n \geq n_0$. Therefore $X_n(\omega) \rightarrow 0$, for all $\omega \in (0, 1]$, i.e. $X_n \xrightarrow{a.s.} 0$.

For any $\varepsilon > 0$ we have

$$\mathbb{P}(X_n \geq \varepsilon) \leq \frac{1}{n}, n \in \mathbb{N},$$

therefore $X_n \xrightarrow{\mathbb{P}} 0$. Note that

$$\mathbb{E}[|X_n|^r] = \frac{e^{rn}}{n} \rightarrow +\infty, r > 0.$$

therefore $X_n \not\xrightarrow{(r)} 0$.

Examples

Let $(\Omega, \mathcal{F}, \mathbb{P})$ is some probability space, such that $\mathbb{P}(A) = \mathbb{P}(\overline{A}) = 0.5$ for some $A \in \mathcal{F}$.

Examples

Let $(\Omega, \mathcal{F}, \mathbb{P})$ is some probability space, such that $\mathbb{P}(A) = \mathbb{P}(\overline{A}) = 0.5$ for some $A \in \mathcal{F}$. Denote

$$X_n(\omega) = \begin{cases} 1, & \text{if } \omega \in A, \\ -1, & \text{if } \omega \in \overline{A} \end{cases}$$

and $X = -X_1$.

Examples

Let $(\Omega, \mathcal{F}, \mathbb{P})$ is some probability space, such that $\mathbb{P}(A) = \mathbb{P}(\overline{A}) = 0.5$ for some $A \in \mathcal{F}$. Denote

$$X_n(\omega) = \begin{cases} 1, & \text{if } \omega \in A, \\ -1, & \text{if } \omega \in \overline{A} \end{cases}$$

and $X = -X_1$. It is not difficult to see that

$$F_n(x) = F(x) = \begin{cases} 0, & \text{if } x < -1, \\ 0.5, & \text{if } x \in [-1, 1), \\ 1, & \text{if } x \geq 1. \end{cases}$$

Examples

Let $(\Omega, \mathcal{F}, \mathbb{P})$ is some probability space, such that $\mathbb{P}(A) = \mathbb{P}(\overline{A}) = 0.5$ for some $A \in \mathcal{F}$. Denote

$$X_n(\omega) = \begin{cases} 1, & \text{if } \omega \in A, \\ -1, & \text{if } \omega \in \overline{A} \end{cases}$$

and $X = -X_1$. It is not difficult to see that

$$F_n(x) = F(x) = \begin{cases} 0, & \text{if } x < -1, \\ 0.5, & \text{if } x \in [-1, 1), \\ 1, & \text{if } x \geq 1. \end{cases}$$

However $|X_n - X| = 2$.

Examples

Let $(\Omega, \mathcal{F}, \mathbb{P})$ is some probability space, such that $\mathbb{P}(A) = \mathbb{P}(\overline{A}) = 0.5$ for some $A \in \mathcal{F}$. Denote

$$X_n(\omega) = \begin{cases} 1, & \text{if } \omega \in A, \\ -1, & \text{if } \omega \in \overline{A} \end{cases}$$

and $X = -X_1$. It is not difficult to see that

$$F_n(x) = F(x) = \begin{cases} 0, & \text{if } x < -1, \\ 0.5, & \text{if } x \in [-1, 1), \\ 1, & \text{if } x \geq 1. \end{cases}$$

However $|X_n - X| = 2$. So $X_n \xrightarrow{D} X$, but $X_n \not\xrightarrow{\mathbb{P}} X$.

The Weak Law of Large Numbers

The Weak Law of Large Numbers

Markov's theorem

Let X_n be a sequence of random variables, each having a finite mean ($\mathbb{E}[X_k] = a_k$) and variance. If

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \text{Var} \left(\sum_{k=1}^n X_k \right) = 0, \text{ then } \frac{1}{n} \sum_{k=1}^n (X_k - a_k) \xrightarrow{\mathbb{P}} 0.$$

The Weak Law of Large Numbers

Markov's theorem

Let X_n be a sequence of random variables, each having a finite mean ($\mathbb{E}[X_k] = a_k$) and variance. If

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \text{Var} \left(\sum_{k=1}^n X_k \right) = 0, \text{ then } \frac{1}{n} \sum_{k=1}^n (X_k - a_k) \xrightarrow{\mathbb{P}} 0.$$

Corollary

If X_n is a sequence of independent and identically distributed random variables, each having a finite mean ($\mathbb{E}[X_1] = a$) and variance, then

$$\frac{1}{n} \sum_{k=1}^n X_k \xrightarrow{\mathbb{P}} a.$$